

Exponential and Log Functions

ECON 441: Introduction to Mathematical Economics
Lecture 10

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Example

Total cost: $C = C(Q)$

Marginal cost: $MC = C'(Q)$

Average cost:

$$AC = \frac{C(Q)}{Q}$$

When is $\frac{dAC}{dQ}$ positive?

Example

Revenue: $R = f(Q)$

Output: $Q = g(L)$

How does revenue change due to a change in labor input L ?

$$\frac{dR}{dL} = \frac{dR}{dQ} \cdot \frac{dQ}{dL}$$

Exponential Functions

The exponential function can be represented as:

$$y = f(t) = b^t \quad (b > 1)$$

where b denotes a fixed base of the exponent.

A more generalized version can be written as:

$$y = ab^{ct}$$

Natural Exponential Function

Natural exponential function: the base is a special mathematical constant called Euler's number $e = 2.71828\dots$

$$y = ae^{rt}$$

Where did this number e come from?

It can be shown:

$$e \equiv \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

Natural Exponential Function

Jacob Bernoulli discovered this constant in 1683 while studying a question about compound interest.

Invest A dollars at a 5% interest rate that compounds yearly. After one year you have $A(1 + 0.05)$; after t years:

$$A(1 + 0.05)^t$$

Natural Exponential Function

After t years: $A(1 + 0.05)^t$. Rewrite this as:

$$A \left(\underbrace{\left(1 + \frac{1}{1/0.05} \right)^{1/0.05}}_{\approx e} \right)^{0.05t} \approx Ae^{0.05t}$$

More generally, with initial principal A_0 , interest rate r , and horizon t :

$$A_t \approx A_0 e^{rt}$$

Logarithmic Function

Since the exponential function is a monotonic function, its inverse exists.

The inverse of the exponential function is called the **log** or **logarithmic** function.

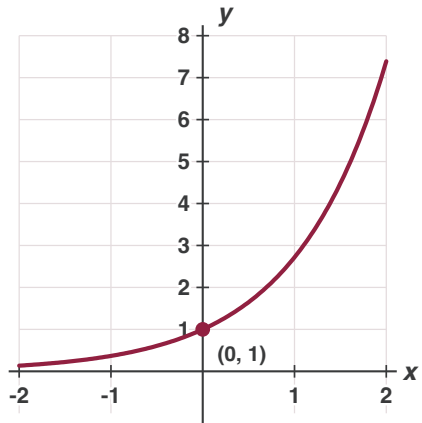
For the exponential function:

$$y = b^t \rightarrow \log_b y = t$$

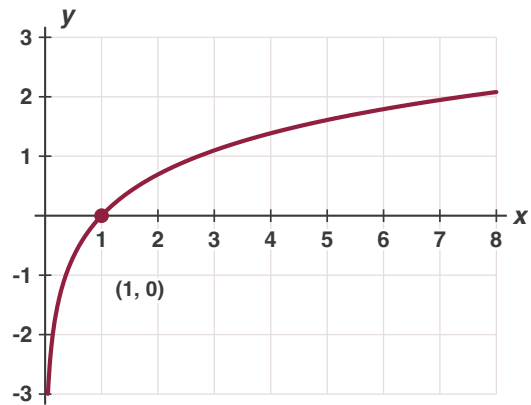
For the natural exponential function:

$$y = e^t \rightarrow \log_e y = \ln y = t$$

$$y = e^x$$



$$y = \ln x$$



Rules for Logarithmic Functions

- $\ln(uv) = \ln u + \ln v$
- $\ln(u/v) = \ln u - \ln v$
- $\ln u^a = a \ln u$

Derivatives of Exponential Functions

Derivative of the exponential function:

$$y = e^t \quad \rightarrow \quad \frac{dy}{dt} = e^t$$

Using the chain rule:

$$y = e^{f(t)} \quad \rightarrow \quad \frac{dy}{dt} = f'(t)e^{f(t)}$$

Derivatives of Logarithmic Functions

Derivative of the log function:

$$\frac{d}{dt} \ln t = \frac{1}{t}$$

Using the chain rule:

$$\frac{d}{dt} \ln f(t) = \frac{f'(t)}{f(t)}$$

Examples

Find the derivatives for the following functions:

1. $y = e^t$

2. $y = \ln t$

3. $y = ae^{rt}$

4. $y = e^{-t}$

5. $y = \ln at$

6. $y = \ln t^c$

Homework Questions

- Exercise 10.5: 1, 3, 7

Textbook reference: 10.5

Notes on Log and Exponential Functions are on the website